Advanced Trading Strategies

1 Securities Markets and Liquidity

1.1 Why Microstructure

Textbook asset pricing assumes every participant is present at once, trades at a single price that reflects a consensus on fundamentals, and pays no costs. In real markets traders arrive at different times, hold different information (insiders, professional market makers, retail investors), and need time to agree on a price. Market microstructure studies how prices are actually formed under these frictions, with three ingredients: **private information**, **heterogeneous participants** (dealers, hedgers, speculators) and the **institutional rules** of the venue.

Motives to Trade: hedging risk, exploiting superior information (speculation), and rebalancing after liquidity shocks.

Liquidity: the extent to which an order can be filled quickly at a price close to the consensus value. Buys push prices up and sells push them down (**price impact**), and the ask exceeds the bid (**spread**); both are measures of illiquidity.

Price Discovery: how quickly and accurately trade prices absorb the information held by market participants. Prices can incorporate information well before it is publicly confirmed, since order flow itself carries information.

Why It Matters: trading costs are first order for asset managers and retail investors, time varying liquidity is priced (an illiquidity premium raises the cost of capital), and prices serve as signals for real investment and as performance benchmarks.

1.2 Market Structure

Venues: regulated exchanges, multilateral trading facilities (alternative trading systems), dealer operated OTC markets, and internalisation by broker dealers.

Liquidity Takers vs Makers: takers demand immediate execution at the best quotes and pay the spread; makers post quotes, wait patiently and earn the spread.

Dealer (quote driven)	Auction (order driven)
Decentralised, intermediated	Centralised, direct
Dealers quote bid and ask and trade on own account	Orders meet in an electronic limit order book
No price priority, bargaining possible, quotes often not visible	Price and time priority, book is transparent
Search costs, inventory risk for dealers	Execution risk for limit orders
Retail segment plus an interdealer segment to share inventory	Call (batch) or continuous trading

Order Types:

- **Market order:** trade a quantity now at the best available prices; fast, but pays for immediacy
- **Limit order:** trade up to a size at a stated worst price; better price, but faces **execution risk** and the risk of being **picked off** when news arrives
- **Stop order:** becomes active once a trigger price is reached; useful to cap losses, but demands liquidity exactly when it is scarce
- **Iceberg (hidden) order:** only part of the size is displayed, which conceals a large order

Liquidity rests on a balance between the two flows: too many limit orders and makers struggle to execute; too few and takers face thin books.

Call Auction: buy orders are ranked by decreasing price (demand) and sell orders by increasing price (supply); a single clearing price equates the two and every executable order trades at that price.

Continuous Limit Order Book (LOB): unexecuted limit orders rest in the book, and each incoming market order is matched against it by price, then time priority. A market order larger than the best level **walks the book**, paying progressively worse prices.

Call vs Continuous Efficiency: if limit prices reflect true valuations, the call auction is Walrasian: everyone valuing the asset above the clearing price ends up holding it and nobody else does, so the allocation is **Pareto efficient**. At the clearing price, executable orders on the long side of the market may be only partly filled, following priority rules. A continuous market usually generates *more* volume, yet some trades occur between traders who would not have traded in a call (a buyer valuing the asset below the eventual clearing price, for example), and early market orders walk a thin book. The cost of immediacy is lost gains from trade, weighed against the convenience of trading at any moment.

1.3 Measuring Trading Costs

Explicit costs are fees, commissions, taxes and settlement charges. The main implicit cost is the spread. With best ask a and best bid b:

$$S = a - b, \qquad m = \frac{a+b}{2}, \qquad s = \frac{a-b}{m}$$

the quoted spread, midprice and relative quoted spread.

Weighted Average Spread: for a market order of size q with average execution prices $\bar{a}(q)$ and $\bar{b}(q)$, $s(q) = (\bar{a}(q) - \bar{b}(q))/m$, which captures depth beyond the best quotes.

Effective Half Spread: using the trade price p, the prevailing mid m and direction $d = +1$ (buy) or -1 (sell),

$$s_e = \frac{d\,(p-m)}{m}$$

Signing Trades (Lee and Ready): a trade above the mid is a buy and below is a sell; at the mid, use the tick test (an uptick is a buy, a downtick a sell). Accuracy is imperfect, roughly 72% to 85% in empirical studies.

Realised Spread: what the liquidity supplier actually keeps once the price has moved, measured against the mid a short time Δ later:

$$s_r = \frac{d_t(p_t - m_{t+\Delta})}{m_t} = \underbrace{\frac{d_t(p_t - m_t)}{m_t}}_{\text{effective}} - \underbrace{\frac{d_t(m_{t+\Delta} - m_t)}{m_t}}_{\text{price impact}}$$

Amihud Illiquidity: daily price impact per unit of trading value, $|\text{Ret}_d|/(\text{Vol}_d \times \text{Price}_d)$.

VWAP: without quote data, VWAP $= \sum_t w_t p_t$ with $w_t = |q_t| / \sum_s |q_s|$, often computed separately for buys and sells.

Empirical Notes:

- The log spread $\ln a - \ln b$ is almost identical to the relative spread for typical spread sizes
- Spreads follow an intraday pattern, typically widest just after the open when uncertainty and information asymmetry peak
- Buyer initiated VWAP sits above the overall VWAP and seller initiated VWAP below it; the gap is a volume weighted cost of immediacy
- **Estimating price impact:** regress mid price changes over short intervals on cumulative signed order flow, $m_{t+h} - m_t = \lambda x_{t,t+h} + e_t$, with x scaled by total volume; a significant $\hat{\lambda}$ is an empirical Kyle lambda

2 Adverse Selection and Inventory Risk

If dealers were risk neutral, competitive, faced no costs and knew as much as their clients, they would quote bid = ask = expected value and the spread would vanish. A positive spread therefore needs a friction:

- **Adverse selection:** some clients are better informed
- **Order processing costs**
- **Inventory holding costs** when dealers are risk averse
- **Rents** from imperfect competition

2.1 Glosten and Milgrom (1985)

Setup: the asset is worth v_H or v_L with equal prior probability. Each period one trader arrives: with probability π an **informed** trader who buys if $v = v_H$ and sells if $v = v_L$; with probability $1 - \pi$ a **noise** trader who buys or sells with probability $1/2$. Competitive, risk neutral dealers earn zero expected profit, so

$$a_t = \mathbb{E}(v \mid \Omega_{t-1}, q_t = +1), \qquad b_t = \mathbb{E}(v \mid \Omega_{t-1}, q_t = -1)$$

The quotes are *regret free*: each is the fair value conditional on the trade that would hit it.

Why Order Flow Is Informative: $\mathbb{P}(q = +1 \mid v_H) = \frac{1+\pi}{2}$ and $\mathbb{P}(q = +1 \mid v_L) = \frac{1-\pi}{2}$, so a buy makes the high value more likely.

First Quotes: with $\mu_0 = (v_H + v_L)/2$, zero profit on a buy requires $\frac{\pi}{2}(a_1 - v_H) + \frac{1-\pi}{2}(a_1 - \mu_0) = 0$, giving

$$a_1 = \mu_0 + \tfrac{\pi}{2}(v_H - v_L), \qquad b_1 = \mu_0 - \tfrac{\pi}{2}(v_H - v_L), \qquad S_1 = \pi(v_H - v_L)$$

The spread exactly offsets expected losses to informed traders with gains from noise traders.

Bayesian Updating: with $\theta_{t-1} = \mathbb{P}(v = v_H \mid \Omega_{t-1})$,

$$\theta_t^+ = \frac{\frac{1+\pi}{2}\theta_{t-1}}{\pi\theta_{t-1} + \frac{1-\pi}{2}}, \qquad \theta_t^- = \frac{\frac{1-\pi}{2}\theta_{t-1}}{\pi(1-\theta_{t-1}) + \frac{1-\pi}{2}}$$

and the general spread is

$$S_t = 2\pi\theta_{t-1}(1-\theta_{t-1})\left[\frac{1}{2\pi\theta_{t-1} + 1 - \pi} + \frac{1}{2\pi(1-\theta_{t-1}) + 1 - \pi}\right](v_H - v_L)$$

Implications:

- The spread increases in the share of informed trading π (the idea behind the PIN measure) and in value uncertainty $v_H - v_L$, and peaks when $\theta_{t-1} = 1/2$
- Quotes move up after buys and down after sells, and the impact is **permanent**: there is no reversal
- Trading itself generates volatility, and prices converge to the true value as trades reveal information

Market Breakdown: if uninformed traders only trade when the price lies within δ of the public value μ, then with $v_H - v_L = 1$ the first period spread π is acceptable only when $\pi \leq 2\delta$. If adverse selection is more severe, dealers infer that any order is informed, quote $a = v_H$ and $b = v_L$, informed traders earn nothing, and trading can vanish entirely. Too much private information can shut a market down.

Costly Information: suppose insiders must pay c to learn v, and dealers expect them to do so with probability ϕ. Replacing π by the effective share $\pi\phi$ gives

$$S = \frac{\pi\phi}{\pi\phi + 1 - \pi}(v_H - v_L), \qquad \text{insider profit} = \frac{1}{2}\frac{1-\pi}{\pi\phi + 1 - \pi}(v_H - v_L) - c$$

The spread rises and the insider's gross gain falls as ϕ rises, so information acquisition is self limiting. An interior equilibrium sets the expected gain equal to c: the more people are expected to become informed, the less it pays to become informed.

PIN (Easley, Kiefer, O'Hara and Paperman): a structural version of this logic. On each day an information event occurs with probability α; informed orders then arrive at Poisson rate μ on one side, while uninformed buys and sells arrive at rate ε each. Fitting daily buy and sell counts by maximum likelihood gives

$$\text{PIN} = \frac{\alpha\mu}{\alpha\mu + 2\varepsilon}$$

the share of order flow that is informed. Less frequently traded stocks tend to show higher PIN and wider spreads. Limitations: the Poisson and one event per day structure, numerical problems for very active stocks, and the risk that PIN captures order imbalance or liquidity shocks rather than information.

2.2 Stoll (1978): Inventory Holding Costs

Setup: no informed traders and no processing costs. A myopic dealer with inventory z_t and cash c_t supplies y_t shares at price p_t ($y_t > 0$ sells) and values the position at the final payoff v, with $\mathbb{E}_t(v) = \mu_t$ and standard deviation σ_v. Mean standard deviation preferences give

$$U = \mathbb{E}_t(w_{t+1}) - \rho\,\mathrm{sd}_t(w_{t+1}) = \mu_t(z_t - y_t) + c_t + p_t y_t - \rho\sigma_v|z_t - y_t|$$

Quotes: $\partial U/\partial y_t = p_t - \mu_t \mp \rho\sigma_v$ depending on the sign of $z_t - y_t$, so in equilibrium

$$a_t = \mu_t + \rho\sigma_v, \qquad b_t = \mu_t - \rho\sigma_v, \qquad S_t = 2\rho\sigma_v$$

The spread is compensation for bearing risk and grows with risk aversion and asset volatility.

Dynamics: a sell order raises the dealer's inventory, so quotes and the mid fall to attract buyers; as inventory normalises the mid drifts back. Inventory effects are **transitory** and reverse gradually.

Order Size Relative to Inventory: with a long inventory $z_t > 0$, a buy order of size $y_t < z_t$ only reduces the dealer's risk, so the dealer sells at $\mu_t - \rho\sigma_v$, the same as the bid, and the spread is zero. Only buys large enough to flip the dealer short are charged $\mu_t + \rho\sigma_v$. Prices are a step function of order size, with the step located at the current inventory.

Cross Sectional Prediction: the inventory premium $\rho\sigma_v$ per unit of position is larger when the asset is riskier and dealers are more risk averse or fewer. This is consistent with much stronger price pressure per unit of inventory in small capitalisation stocks.

Estimating Inventory Effects: with dealer level data (signed trade size q_t, trade sign d_t, inventory z_t), a regression of the form

$$\Delta p_t = \theta_0 q_t + \theta_1 \Delta q_t + \theta_2 \Delta z_t + \gamma \Delta d_t + \varepsilon_t$$

separates the channels: $\theta_0 > 0$ is the information content of order flow; θ_1 mixes inventory effects (positive) and quantity discounts (negative); $\theta_2 < 0$ signals inventory control (after selling, the dealer raises quotes to attract sellers); γ captures the fixed, order processing part of the spread. A regression $z_t - z_{t-1} = a + \rho z_{t-1} + e_t$ with $\hat{\rho} < 0$ indicates mean reverting inventories. For a dealer at target inventory, the per unit cost of a buy of size q is roughly $p_t - m_t = (\theta_0 + \theta_1)q + \gamma$, and the total cost is this times q.

2.3 Price Impact of a Buy Order by Source

Component	Short run	Medium run	Long run
Adverse selection	> 0	> 0	= short run
Inventory risk	> 0	> 0, declining	0
Order processing	> 0	0	0

The spread thus combines an information component, an inventory component and a processing component. Observed negative autocorrelation of returns points to the latter two, which pure adverse selection cannot produce.

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3 Trade Size and Market Depth

A market can be cheap for small trades yet expensive for large ones; it then lacks **depth**. Large orders are more likely to be informed, expose dealers to more inventory risk, and invite market power. A deep market is one where large orders move prices only slightly more than small ones.

3.1 Kyle (1985)

Players: an insider observes $v \sim \mathcal{N}(\mu, \sigma_v^2)$ and submits $x = X(v)$; noise traders submit $u \sim \mathcal{N}(0, \sigma_u^2)$ independent of v; in a batch auction, competitive risk neutral market makers see only the net flow $q = x + u$ and set $p = \mathbb{E}(v \mid q)$.

Market Maker Inference: conjecture a linear strategy $x = \beta(v - \mu)$. Then $q \sim \mathcal{N}(0, \beta^2\sigma_v^2 + \sigma_u^2)$ and, by the normal projection formula,

$$p(q) = \mu + \lambda q, \qquad \lambda = \frac{\mathrm{Cov}(v,q)}{\mathrm{Var}(q)} = \frac{\beta\sigma_v^2}{\beta^2\sigma_v^2 + \sigma_u^2}$$

λ is the price impact (Kyle's lambda) and $1/\lambda$ is **depth**. Order flow informativeness rises with σ_v^2, falls with σ_u^2, and is non monotonic in β (aggression adds information but also inflates the order). The implied spread grows with size, $S(q) = 2\lambda q$.

Insider's Problem: taking λ as given, $\max_x \mathbb{E}[x(v - \mu - \lambda x)]$ gives

$$x = \frac{v - \mu}{2\lambda} \;\Rightarrow\; \beta = \frac{1}{2\lambda}$$

Equilibrium: solving both conditions jointly,

$$\lambda = \frac{\sigma_v}{2\sigma_u}, \qquad \beta = \frac{\sigma_u}{\sigma_v}, \qquad \text{depth} = \frac{2\sigma_u}{\sigma_v}$$

Depth rises with noise trading (better camouflage) and falls with fundamental uncertainty. The insider's expected profit is $\beta\sigma_v^2(1 - \lambda\beta) = \sigma_u\sigma_v/2$, paid for by the noise traders.

Information Revelation: splitting the total variance σ_v^2,

$$\mathrm{Var}(p - \mu) = \frac{\beta^2\sigma_v^2}{\beta^2\sigma_v^2 + \sigma_u^2}\sigma_v^2, \qquad \mathrm{Var}(v - p) = \frac{\sigma_u^2}{\beta^2\sigma_v^2 + \sigma_u^2}\sigma_v^2$$

In equilibrium each equals $\sigma_v^2/2$: exactly half of the private information is impounded in the price.

Extensions:

- **Several rounds:** the insider trades gradually to avoid revealing information too quickly, which slows price discovery
- **Several insiders** (Holden and Subrahmanyam): competition dissipates their advantage quickly and depth rises
- **Imperfectly competitive market makers** (Kyle 1989): market power reduces depth

Imperfectly Informed Insider: if the insider sees only $s = v + \eta$ with independent noise $\eta \sim \mathcal{N}(0, \sigma_\eta^2)$, the same steps give

$$\lambda = \frac{\sigma_v^2}{2\sigma_u\sqrt{\sigma_v^2 + \sigma_\eta^2}}, \qquad \beta = \frac{\sigma_u}{\sqrt{\sigma_v^2 + \sigma_\eta^2}}$$

Both fall as the signal gets noisier: order flow is less informative, so dealers offer more depth, and the insider trades more cautiously because $\mathbb{E}(v \mid s)$ moves less with s. With $\sigma_\eta = 0$ we recover the baseline.

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4 Limit Order Markets

Static models (Glosten 1994, Seppi 1997, Biais, Martimort and Rochet 2000) study the liquidity of the book at a point in time. **Dynamic models** (Parlour 1998, Foucault 1999, Rosu 2009, Goettler, Parlour and Rajan) study the choice between market and limit orders and how the state of the book and the order flow interact.

4.1 Glosten (1994)

Setup: the asset pays $v = \mu + \epsilon$ with $\mathbb{E}(\epsilon) = 0$. Prices lie on a grid $B_n < \cdots < B_1 < \mathbb{E}(v) < A_1 < \cdots < A_n$ with tick size Δ, and execution is **discriminatory**: each limit order trades at its own price, so a large market order walks the book. In period 1 many limit order traders build the book under time priority with free entry; in period 2 a single market order of signed size q arrives, from an informed or a liquidity trader. Since informed traders buy on good news, q and v are positively related.

Marginal Profit: with y_k shares at A_k and cumulative depth $Y_k = \sum_{j \le k} y_j$, the last share offered at A_k earns

$$\Pi_k(Y_k) = \mathbb{P}(q \ge Y_k)\big(A_k - \mathbb{E}(v \mid q \ge Y_k)\big)$$

The **upper tail expectation** $\mathbb{E}(v \mid q \ge Y)$ rises with Y: a share deep in the queue is executed only by large orders, which are more likely informed. Hence Π_k eventually decreases in depth.

Competitive Equilibrium: no profitable entry and no profitable exit, i.e.

$$A_k = \mathbb{E}(v \mid q \ge Y_k^*) \text{ if } Y_k^* > Y_{k-1}^*, \qquad A_k < \mathbb{E}(v \mid q \ge Y_{k-1}^*) \text{ if } Y_k^* = Y_{k-1}^*$$

Two Size Illustration: let $v = \mu \pm \sigma$ equally likely, the trader be informed with probability π, and an uninformed trader buy or sell a small size q_S (probability ϕ) or a large size q_L. With zero tick size the informed trader always trades at least q_L, and the ask schedule is

$$A(Y) = \begin{cases} \mu + \pi\sigma, & Y \le q_S \\ \mu + \dfrac{\pi}{\pi + (1-\pi)(1-\phi)}\sigma, & q_S < Y \le q_L \\ \mu + \sigma, & Y > q_L \end{cases}$$

using $\mathbb{E}(v \mid q \ge Y) = \mu + \mathbb{P}(I \mid q \ge Y)\sigma$.

Implications: the small trade spread is $2\pi\sigma$; a higher π raises costs for all sizes; small uninformed traders pay a spread even though informed traders never trade small, a **cross subsidy** between order sizes.

Limitations: one period only, no choice between market and limit orders, and the zero profit condition is imposed rather than derived from strategic behaviour.

Tick Size as a Rent: with a discrete grid, the first share at A_k executes only if $q \ge Y_{k-1}$, and equilibrium at the previous level sets $A_{k-1} = \mathbb{E}(v \mid q \ge Y_{k-1})$. Its expected profit is therefore

$$\Pi_k(Y_{k-1}) = \mathbb{P}(q \ge Y_{k-1})\,\Delta$$

proportional to the tick. Deeper shares at the same price earn less (lower execution probability and a worse upper tail expectation). The tick protects liquidity suppliers because undercutting must be by at least Δ; with $\Delta = 0$ their profit is zero. **Halving the tick** narrows the spread and spreads depth over more price levels (fewer shares at each old grid price), but leaves the *cumulative* depth up to any old grid price unchanged, since that is pinned down by the zero profit condition on the marginal share.

Maker Taker Fees (Foucault 1999 setting): when the platform charges f_{mo} per share to market orders and f_{lo} to executed limit orders, equilibrium quotes absorb the fees. The quoted spread *increases* with the maker fee, which limit order traders pass into their quotes, and *decreases* with the taker fee, since a costlier market order forces limit orders to be priced more aggressively to remain attractive. Quoted spreads alone can therefore mislead when fee structures differ across venues; the cost including fees is what matters.

4.2 Liquidity Risk and Asset Prices

Acharya and Pedersen (2005) add a relative illiquidity cost c to a CAPM written in net returns. With λ_t the market risk premium net of illiquidity costs,

$$\mathbb{E}_t(r_{t+1}^i) = r_f + \mathbb{E}_t(c_{t+1}^i) + \lambda_t\big(\beta_1 + \beta_2 - \beta_3 - \beta_4\big)$$

where each β is a covariance divided by $\mathrm{Var}_t(r^M - c^M)$:

- $\beta_1 \propto \mathrm{Cov}(r^i, r^M)$: the usual market beta
- $\beta_2 \propto \mathrm{Cov}(c^i, c^M)$: **commonality**, an asset that becomes illiquid when the market does is riskier
- $\beta_3 \propto \mathrm{Cov}(r^i, c^M)$: an asset whose return holds up when market liquidity dries up is a hedge, so it lowers required returns
- $\beta_4 \propto \mathrm{Cov}(c^i, r^M)$: an asset that stays liquid in down markets is valuable, so this term enters negatively

The **illiquidity premium** is the part of expected returns attributable to $\mathbb{E}(c^i)$ and the liquidity betas. Empirically, the covariance of an asset's illiquidity with the market return (β_4) carries the largest share of the liquidity risk premium. Caveats: an Amihud type proxy for c, strong collinearity between the betas, and identification that rests on the model's structure.

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5 Technology and Trading

5.1 Algorithmic and High Frequency Trading

Limit orders risk being **picked off** when news arrives before they can be revised. Remedies include **pegged** orders that reprice automatically and frequent **monitoring**, which is costly. Lower latency and computerisation have collapsed monitoring costs, giving rise to **algorithmic trading** (smart routing across venues, splitting large orders) and its subset, **high frequency trading** (HFT).

HFT Hallmarks: extreme speed, sophisticated algorithms, co-location, very high order to cancel ratios and near zero end of day inventory. HFT is roughly half of

US equity volume, and its fixed costs (co-location, data feeds, infrastructure, staff) are so large that only very high volume is profitable.

Strategies:

- **Cross market arbitrage:** trade on stale quotes at a slower venue; keeps prices aligned but fuels a costly latency arms race
- **Order book signals:** trade ahead of predicted pressure from book imbalance or detected order splitting
- **Market making:** quote both sides and requote constantly; profit $\approx$ spread $\times$ volume less adverse selection, inventory and technology costs; quotes widen or vanish when volatility spikes
- **Rebate arbitrage:** under maker taker pricing, makers earn a rebate and takers pay a fee, so passive round trips can profit from rebates alone
- **Directional and statistical arbitrage** at high frequency

Empirically HFTs act mostly as market makers.

Liquidity Effects: if algorithms mainly help liquidity demanders, spreads should widen; if they help suppliers, spreads should narrow. Hendershott, Jones and Menkveld (2011) use the staggered introduction of automated quote dissemination at the NYSE as an instrument and proxy algorithmic activity by electronic messages per unit of volume; they find narrower spreads, driven by lower adverse selection, and more informative quotes.

Benefits vs Risks:

- Benefits: more liquidity, lower short term volatility, faster price discovery, cheaper retail execution, possible welfare gains
- Risks: adverse selection imposed on slower traders, fast moves away from fundamentals, and propagation of shocks across assets, as in the May 2010 Flash Crash where a large futures sale triggered a brief collapse that largely reversed within the day

5.2 Decentralised Exchanges

A **decentralised exchange** (DEX) runs on a blockchain through smart contracts. Most are **automated market makers** (AMMs): a fixed, public pricing rule acts as the single counterparty, so no order matching is needed, and users keep custody of their assets.

Centralised (CEX)	Decentralised (DEX)
Limit order book	Automated market maker
Proprietary, opaque code	Open source, transparent
Exchange holds custody	User holds custody
Counterparty and fraud risk	Smart contract risk

Constant Product AMM (Uniswap v2): liquidity providers (LPs) deposit two tokens with quantities x and y, fixing $K = xy$. Any trade must preserve K: removing Δx of token X requires adding

$$\Delta y = \frac{K}{x - \Delta x} - y = \frac{y\,\Delta x}{x - \Delta x}$$

The marginal price is y/x, and the average price paid rises with size, so **price impact is built in**. A fee (e.g. 30 bp) is charged on each trade.

Illustration: a pool of 20 BTC and 400 ETH ($K = 8000$) prices BTC at 20 ETH, consistent with BTC at \$50,000 and ETH at \$2,500. Buying 4 BTC leaves $x = 16$, so $y = 500$ and the buyer pays 100 ETH, an average of 25 ETH per BTC. At market prices the pool receives \$250k and gives up \$200k; the \$50k gain is compensation for providing liquidity and bearing adverse selection.

Adverse Selection of AMMs: the curve cannot update when outside prices move. If BTC jumps to \$60,000 (24 ETH), an arbitrageur buys from the pool until $y/x = 24$, i.e. $x = \sqrt{8000/24} \approx 18.26$: about 1.74 BTC worth \$104.6k for 38.2 ETH worth \$95.4k, a \$9.1k gain at the LPs' expense (sniping, or loss versus rebalancing).

Why AMMs Survive: LP returns increase with noise flow (fees) and decrease with adverse selection. Although K and the fee are fixed, the pool size adjusts: a high fee yield attracts deposits, raising K and lowering price impact, while heavy sniping triggers withdrawals. Empirically, pools with a low sniping to noise ratio see inflows and those with a high ratio see outflows.

Concentrated Liquidity (Uniswap v3): each LP chooses a price range $[p_a, p_b]$ on which their position is active, with reserves following

$$\left(x + \frac{L}{\sqrt{p_b}}\right)\left(y + L\sqrt{p_a}\right) = L^2$$

Narrow ranges near the market price earn more fees per unit of capital but are exposed to more adverse selection and fall out of range more often. LPs can widen the range or **reposition** frequently, but each update costs gas. Cheap scaling layers (rollups, sidechains) lower these costs, and liquidity is indeed more concentrated near the market price where repositioning is cheap.

Ticks and Reserves in v3: prices live on ticks $p_i = 1.0001^i$, one basis point apart, with p the price of token X in units of Y. On a tick range $[p_i, p_{i+l}]$ with liquidity L_i (the sum of all positions active there), and with z_i the current price clipped to the range,

$$x_i = \frac{L_i}{\sqrt{z_i}} - \frac{L_i}{\sqrt{p_{i+l}}}, \qquad y_i = L_i\big(\sqrt{z_i} - \sqrt{p_i}\big)$$

Ranges above the current price hold only X, ranges below hold only Y, and only the range containing the current price holds both. A deposit of one token into a range around the current price pins down L and hence the matching amount of the other token. Raw on chain prices must be rescaled by $10^{d_y - d_x}$ for token decimals.

Swaps and Slippage: within one range the curve $(x_i + \Delta x_i + L_i/\sqrt{p_{i+l}})(y_i + \Delta y_i + L_i\sqrt{p_i}) = L_i^2$ gives the output for a given input; a large trade crosses ranges and is computed range by range. The average execution price relative to the pre trade price defines **slippage**, which shrinks as more liquidity is concentrated around the current price.

LP Decisions: which pool to supply (fees vs adverse selection), how wide a range to choose, and how often to reposition given transaction costs.

Takeaway: technology changes speed and form, but the economics are unchanged: liquidity provision is still a trade off between spread or fee income, adverse selection and inventory risk.

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6 Building and Backtesting Algorithmic Strategies

6.1 Algorithmic Trading

In algorithmic trading, computers read market data and send, manage and cancel orders without immediate human intervention, often within milliseconds. Algorithms now account for the majority of US equity volume and an even larger share of futures volume.

Strengths: speed, consistency without emotional bias, the ability to test rules on historical data, handling of many assets at once, and lower execution costs.
Weaknesses: expensive infrastructure, technical complexity, fragility to regime changes, correlated behaviour across similar algorithms (systemic risk), and dependence on technology. A single faulty software deployment can generate millions of unintended orders and ruinous losses within an hour, which is why testing, kill switches and human oversight remain essential.

Strategy Families: directional strategies, execution algorithms for large institutional orders, market making, and statistical arbitrage such as pairs trading.

Directional Signals:

- **Momentum:** past winners keep winning over roughly 3 to 12 months (Jegadeesh and Titman), attributed to behavioural underreaction; signals from past returns or moving average crossovers
- **Mean reversion:** at very short horizons from bid ask bounce, order flow shocks and temporary liquidity imbalances (the inventory logic of Stoll); at 3 to 5 year horizons from overreaction (De Bondt and Thaler); signals from deviations from moving averages or z scores of recent returns

6.2 The Research Pipeline

1. **Model:** keep it as simple as possible but no simpler, and give every rule an economic rationale
2. **Data:** clean and construct variables
3. **Estimation:** choose a model class
4. **Validation:** out of sample and walk forward testing
5. **Backtest:** translate forecasts into trades and measure performance
6. **Attribution:** check whether returns are alpha or factor exposure
7. **Paper trading:** run live on simulated capital
8. **Production:** monitoring and risk management

Microstructure Features for Short Horizon Returns: for a next minute mid quote return R_{t+1}, natural predictors are the relative quoted spread (bid ask bounce), traded volume and realised volatility (temporary price pressure, which tends to revert), **order imbalance** (signed volume as a share of total volume) and **depth imbalance** (ask size $-$ bid size)/(ask size $+$ bid size) at the best quotes. Order imbalance and depth imbalance are often strongly correlated, so including both in one linear regression invites multicollinearity; check the correlation matrix first.

Data Hygiene: keep regular trading hours, positive quotes and sizes, match each trade with the prevailing quote on the same venue, aggregate to a fixed frequency (e.g. one minute, with returns from closing mid quotes), and winsorise variables at the 1st and 99th percentiles. Decide explicitly between mid quote and trade price returns and between time and volume weighted averages.

6.3 Estimation and Validation

Models: OLS or ARMA for continuous returns, logistic regression for the sign of returns, and tree ensembles (random forests, gradient boosting such as XGBoost) for nonlinear effects. Standardise features for linear models, and *fit the scaler on the training data only.*

Tree Ensemble Settings: intraday returns are very noisy, so ensembles need regularisation. In a random forest, many trees and a large minimum leaf size limit overfitting; in gradient boosting, shallow trees, a small learning rate, a sensible number of rounds and a large minimum child weight play the same role. Feature importance should be compared across refits and against the size of linear coefficients.

Out of Sample: split chronologically (e.g. first 80% train, last 20% test) and never shuffle a time series. Use a separate validation block when tuning hyperparameters.

Walk Forward: repeatedly fit on a window and predict the next block, then roll forward; for instance, train on the last 10 trading days, predict the next day, refit daily. Windows may be **rolling** (fixed length, adapts to change) or **expanding** (all history, more stable). Store coefficients from each refit to see how stable the relationships are.

Forecast Metrics:

$$\mathrm{RMSE} = \sqrt{\tfrac{1}{N}\textstyle\sum_t (y_t - \hat{y}_t)^2}, \qquad \mathrm{MAE} = \tfrac{1}{N}\textstyle\sum_t |y_t - \hat{y}_t|, \qquad \rho = \mathrm{Corr}(y, \hat{y})$$

together with R^2. RMSE penalises large errors and is sensitive to outliers; MAE is more robust. Out of sample R^2 for minute returns is typically tiny or negative, which is not by itself fatal: what matters is economic value.

Alternative Data: text such as headlines can be turned into a numeric signal with a finance tuned language model (e.g. FinBERT), mapping labels to $+1, 0, -1$ and averaging per day. It must be timestamped carefully and merged only with data that came *after* its release.

6.4 Backtesting

From Forecast to Position: a typical rule takes $s_t = +1$ when the predicted return is positive and in the top quartile of returns, $s_t = -1$ when it is negative and in the bottom quartile, and $s_t = 0$ otherwise. The strategy return is $s_t R_t$.

Performance Metrics:

- Cumulative return $\prod_t (1 + s_t R_t) - 1$
- Sharpe ratio μ_s/σ_s of strategy returns (state the frequency; annualise by $\sqrt{\text{periods per year}}$)
- Hit rate: share of nonzero signals with the correct direction
- Maximum drawdown: $\max_t(\text{peak equity} - \text{equity}_t)/\text{peak equity}$

- Turnover: $\frac{1}{T}\sum_t |s_t - s_{t-1}|/2$, so a switch from fully long to fully short counts as one
- t statistic of the mean return, $\mu_s/(\sigma_s/\sqrt{N})$; $|t| > 1.96$ is significant at 5% in large samples

Pitfalls:

- **Look ahead bias:** using information not available at decision time (end of day prices for intraday decisions, revised macro data, models fitted on future observations)
- **Survivorship bias:** testing only on assets or funds that still exist
- **Overfitting and data snooping:** testing 100 rules at the 5% level yields about 5 false discoveries; warning signs are many parameters, long feature lists, near perfect backtests and no economic story
- **Ignoring costs:** spreads, fees, slippage and impact; a rule earning 2 bp per trade that pays 5 bp in costs loses money, and high turnover strategies are hit hardest

Remedies are disciplined data splits, penalties for complexity, economic priors and **stress tests** across bull and bear markets, volatility regimes, rate cycles and crises.

Is It Tradeable? Significant raw returns are not enough. Regress strategy returns on factor returns,

$$r_t^{\text{strat}} = \alpha + \sum_k \beta_k F_{k,t} + \epsilon_t$$

and require a significant α after costs. Then paper trade on live data, comparing realised fills, slippage and the profit and loss distribution with the backtest.

In Production: monitor performance, signals and execution (fills, latency, rejections) with automated alerts, and control risk with value at risk, drawdown limits, position, leverage and concentration limits, and automatic stop losses.

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7 Optimal Consumption and Investment

The **Merton problem**: an agent with initial wealth x splits it over time between *consumption* and *investment* in a riskless bond and a risky asset, so as to maximise expected (discounted) utility of consumption. The base case has a finite horizon with a bequest or an infinite horizon, discrete or continuous time, constant parameters in a complete market, and CRRA or log utility. Extensions include stochastic volatility or rates, transaction costs, portfolio constraints and habit formation. Two questions recur:

1. Is the problem **well posed**, i.e. is the value finite?
2. If so, what are the optimal consumption and investment rules?

7.1 Utility Functions

CRRA (Power): $U(x) = \dfrac{x^{1-R}}{1-R}$, $R \in (0,\infty)\setminus\{1\}$, with $U' = x^{-R} > 0$ and $U'' = -Rx^{-R-1} < 0$.

$$\text{ARA} = -\frac{U''}{U'} = \frac{R}{x}, \qquad \text{RRA} = -\frac{xU''}{U'} = R$$

so relative risk aversion is constant. For $R < 1$ utility is positive; for $R > 1$ it is negative.

Log ($R = 1$): $U''/U' = -1/x$ gives $U(x) = A\ln x + B$, the limit of the CRRA family.

Exponential (CARA): $U(x) = -e^{-\gamma x}$ with constant absolute risk aversion γ (used in Section 12).

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8 Discrete Time

8.1 One Period Model

Market: a bond $S_1^0 = S_0^0(1+r)$ and a stock $S_1^1 = S_0^1 Z$ with $Z \geq 0$ random and typically $\mathbb{E}(Z) > 1 + r$. No arbitrage requires

$$\mathbb{P}(Z < 1+r) > 0 \quad \text{and} \quad \mathbb{P}(Z > 1+r) > 0$$

Controls and Wealth: consumption c at time 0 and ϑ shares held over $(0,1)$:

$$X_1 = (x - c - \vartheta S_0^1)(1+r) + \vartheta S_0^1 Z = (x-c)(1+r) + \vartheta S_0^1\big(Z - (1+r)\big)$$

We maximise $\mathbb{E}[U_0(c) + U_1(X_1)]$ with U_0, U_1 increasing and concave.

Admissibility: with $a \leq Z \leq b$ tight and $0 \leq a < 1 + r < b \leq \infty$, nonnegative wealth in every state requires, for $y = (x-c)(1+r)$,

$$\vartheta \in I_y = \left[\frac{-y}{S_0^1\big(b-(1+r)\big)},\ \frac{y}{S_0^1\big((1+r)-a\big)}\right]$$

If $a = 0$ then $\vartheta S_0^1 \leq x - c$: borrowing to buy shares is ruled out, because a total loss would leave negative wealth and utility $-\infty$. If $b = \infty$ then $\vartheta \geq 0$: no short selling.

Reduction: define $\tilde{V}_1(\vartheta; y) = \mathbb{E}[U_1(y + \vartheta S_0^1(Z - (1+r)))]$ and $\tilde{V}_1^*(y) = \sup_{\vartheta \in I_y} \tilde{V}_1(\vartheta; y)$. Then

$$V(x) = \sup_{c \in [0,x]} \Big\{U_0(c) + \tilde{V}_1^*\big((x-c)(1+r)\big)\Big\}$$

- **Lemma 1:** $\vartheta \mapsto \tilde{V}_1(\vartheta; y)$ is concave on I_y (concavity of U_1); since $\tilde{V}_1(0; y) = U_1(y) < \infty$, it never equals $+\infty$
- **Lemma 2:** $\tilde{V}_1^*$ is concave and increasing on $[0,\infty)$ (proof via ϵ optimal strategies, as I_y scales linearly in y)
- **Lemma 3:** V is concave in x

At an interior optimum, $U_0'(c^*) = (1+r)(\tilde{V}_1^*)'\big((x-c^*)(1+r)\big)$: the marginal utility of consuming today equals that of saving.

8.2 Multi Period Model and Dynamic Programming

Wealth: with iid returns Z_k, adapted consumption C and holdings ϑ,

$$X_k = (X_{k-1} - C_{k-1} - \vartheta_{k-1} S_{k-1}^1)(1+r) + \vartheta_{k-1} S_{k-1}^1 Z_k, \qquad X_0 = x$$

and we maximise $\mathbb{E}\big[\sum_{j=0}^{N-1} U_j(C_j) + U_N(X_N)\big]$ subject to $X \geq 0$.

Backward Induction: by the tower property, the last period problem $\sup \mathbb{E}[U_{N-1}(C_{N-1}) + U_N(X_N) \mid \mathcal{F}_{N-1}]$ is a one period problem whose value $U_{N-1}(X_{N-1})$ is again concave. It replaces the last two terms, leaving an $(N-1)$ period problem of the same form. Repeating reduces the problem to a sequence of one period problems.

Bellman Equation: for a controlled Markov chain $X_{n+1} = F(n, X_n, U_n, Z_{n+1})$ with iid innovations and cost $\mathbb{E}[\sum_{n<N} K(n, X_n, U_n) + \xi(X_N)]$ to minimise,

$$V_N(x) = \xi(x), \qquad V_k(x) = \inf_u \Big\{K(k,x,u) + \mathbb{E}\big[V_{k+1}(F(k,x,u,Z))\big]\Big\}$$

The optimal control is Markov, $u^*(k,x)$. For utility maximisation, replace inf by sup.

Optimal Stopping as Dynamic Programming: when the only decision is whether to stop and collect a reward $g(X_k)$, the Bellman equation becomes

$$V_N(x) = g(x), \qquad V_k(x) = \max\Big\{g(x),\ \mathbb{E}\big[V_{k+1}(X_{k+1}) \mid X_k = x\big]\Big\}$$

so we stop as soon as the reward beats the continuation value. For example, with up to N independent Uniform$(0,1)$ draws where each draw may be kept or discarded, the value with n draws left solves $v_1 = \frac{1}{2}$ and $v_{n+1} = \mathbb{E}\max(U, v_n) = \frac{1+v_n^2}{2}$, giving $0.5, 0.625, 0.695, 0.742, \ldots$ The acceptance threshold is the continuation value, so the agent is pickier early on and less demanding near the end.

8.3 CRRA: One Period

With $U_j(c) = A_j \frac{c^{1-R}}{1-R}$, scale the risky position by post consumption wealth: $\vartheta S_0^1 = \lambda y$. Then

$$\tilde{V}_1^*(y) = \frac{\kappa A_1}{1-R}\, y^{1-R}, \qquad \frac{\kappa}{1-R} = \sup_{\lambda \in \Lambda} \frac{\mathbb{E}\big[(1 + \lambda(Z - (1+r)))^{1-R}\big]}{1-R}$$

with $\Lambda = \big[\frac{-1}{b-(1+r)}, \frac{1}{(1+r)-a}\big]$, so $\kappa = \mathbb{E}[(1 + \lambda^*(Z-(1+r)))^{1-R}]$ does not depend on wealth.

Optimal Rules: the first order condition $A_0 c^{-R} = \kappa A_1 (1+r)^{1-R}(x-c)^{-R}$ gives

$$c^* = \alpha^* x, \qquad (\alpha^*)^{-1} = 1 + \left(\frac{\kappa A_1}{A_0}\right)^{1/R} (1+r)^{1/R-1}$$

and $\vartheta^* S_0^1 = \lambda^*(x - c^*)(1+r)$: consume a constant fraction of wealth and invest a constant fraction of what remains.

Value:

$$V(x) = \Xi \frac{x^{1-R}}{1-R}, \qquad \Xi = \left(A_0^{1/R} + (\kappa A_1)^{1/R}(1+r)^{1/R-1}\right)^R$$

The value function inherits the CRRA form with the same R, which is what makes induction work.

Well Posedness Conditions: beyond no arbitrage, it is convenient to assume $Z > 0$ and $\mathbb{E}[(Z - (1+r))^+] > 0$, so that investing in the stock has some upside and the portfolio problem has a finite, attained optimum. With a general fraction ϕ of post consumption wealth in the stock, terminal wealth is $(x-c)\big(\phi Z + (1-\phi)(1+r)\big)$, which makes the separation into a consumption and a portfolio problem transparent.

8.4 CRRA: Many Periods and Infinite Horizon

N Periods: guess $V_k(x) = \Xi_k \frac{x^{1-R}}{1-R}$, true at $k = N$ with $\Xi_N = A_N$. Applying the one period result with A_1 replaced by Ξ_k,

$$\Xi_{k-1} = \left(A_{k-1}^{1/R} + (\kappa \Xi_k)^{1/R}(1+r)^{1/R-1}\right)^R$$

$$(\alpha_{k-1}^*)^{-1} = 1 + \left(\frac{\kappa \Xi_k}{A_{k-1}}\right)^{1/R} (1+r)^{1/R-1}$$

The investment fraction λ^* is the same at every date; only the consumption fraction varies with time to maturity.

Infinite Horizon: maximise $\mathbb{E}\big[\sum_{k\geq 0} A\beta^k \frac{C_k^{1-R}}{1-R}\big]$. Trying $V_k(x) = \Xi \beta^k \frac{x^{1-R}}{1-R}$ in the recursion gives $\Xi^{1/R}(1-\tilde{\theta}) = A^{1/R}$ with

$$\tilde{\theta} = \kappa^{1/R}\beta^{1/R}(1+r)^{1/R-1}, \qquad \Xi = A(1-\tilde{\theta})^{-R}, \qquad \alpha^* = 1 - \tilde{\theta}$$

Well posedness: $\tilde{\theta} < 1$, equivalently $\beta\,\kappa\,(1+r)^{1-R} < 1$, which is exactly $\alpha^* > 0$. If it fails, for $R < 1$ some strategy gives $+\infty$, and for $R > 1$ every strategy gives $-\infty$.

8.5 Log Utility

With $U_j = A_j \ln c + B_j$ and the fraction ϕ of post consumption wealth in the stock, the objective **separates** into a consumption part and a portfolio part. Guessing $v_k(x) = \alpha_k \ln x + \beta_k$ with $\alpha_N = A_N$:

$$C_k^* = \frac{A_k}{\alpha_k} X_k, \qquad \alpha_k = \sum_{j=k}^{N} A_j, \qquad \phi^* = \arg\max_{\phi} \mathbb{E}\big[\ln\big(1 + r + \phi(Z - (1+r))\big)\big]$$

The portfolio rule is myopic and constant in time (the growth optimal portfolio). In one period, $c^* = \frac{A_0}{A_0 + A_1} x$ and $V(x) = (A_0 + A_1)\ln x + D$, consistent with the envelope relation $V'(x) = U_0'(c^*(x))$. With weights $A_j = A\rho^j$, consumption becomes a constant fraction $1-\rho$ of wealth once the bequest weight is $A\rho^N/(1-\rho)$, which is also the infinite horizon rule.

8.6 Verification (Infinite Horizon)

A candidate from the recursion must still be proved optimal. Let

$$M_k = \sum_{j=0}^{k-1} A\beta^j \frac{C_j^{1-R}}{1-R} + \beta^k V(X_k), \qquad V(y) = \Xi \frac{y^{1-R}}{1-R}$$

If (1) M is a supermartingale for every admissible strategy, (2) M is a martingale for the candidate, (3) $\liminf_k \beta^k \mathbb{E}[V(X_k)] \geq 0$ for every strategy (**transversality**, automatic when $R < 1$), and (4) $\lim_k \beta^k \mathbb{E}[V(X_k^*)] = 0$ for the candidate, then V is the value function and the candidate is optimal. The proof: $V(x) = M_0 \geq \mathbb{E}[M_k]$, let $k \to \infty$ using monotone convergence and (3) to get $V(x) \geq J$ for any strategy, with equality for the candidate by (2) and (4).

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9 Continuous Time

9.1 Model and Wealth

Bond $\mathrm{d}S_t^0 = rS_t^0\,\mathrm{d}t$; stock $\mathrm{d}S_t/S_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$; Sharpe ratio $\varrho = (\mu - r)/\sigma$. With π_t the *fraction* of wealth in the stock and C_t the consumption rate,

$$\mathrm{d}X_t = X_t\pi_t\sigma\,\mathrm{d}W_t + \big(X_t[r + \pi_t(\mu - r)] - C_t\big)\,\mathrm{d}t, \qquad X_0 = x$$

A strategy is admissible if $X \geq 0$; after hitting zero, wealth, investment and consumption stay at zero. Objectives are $\mathbb{E}[\int_0^T U(s,C_s)\,\mathrm{d}s + B(X_T)]$ or the perpetual $\mathbb{E}[\int_0^\infty U(s,C_s)\,\mathrm{d}s]$ with $U(s,c) = e^{-\delta s}\frac{c^{1-R}}{1-R}$.

9.2 Martingale Optimality Principle and HJB

For $\mathrm{d}X = a(t,X,U)\,\mathrm{d}W + b(t,X,U)\,\mathrm{d}t$ and a one signed reward $J(x,U) = \mathbb{E}\int_0^\infty K(s,X_s,U_s)\,\mathrm{d}s$, set

$$M_t^U = \int_0^t K(s,X_s,U_s)\,\mathrm{d}s + V(t,X_t)$$

Theorem: if $V \in C^{1,2}$ and U^* satisfy (i) M^U is a supermartingale for every admissible U, (ii) M^{U^*} is a martingale, (iii) $\liminf_t \mathbb{E}[V(t,X_t^U)] \geq 0$ for all U, and (iv) $\lim_t \mathbb{E}[V(t,X_t^{U^*})] = 0$, then $\sup_U J(x,U) = V(0,x)$ and U^* is optimal.

HJB Equation: by Itô, $\mathrm{d}M = (K + \mathcal{L}V)\,\mathrm{d}t + aV_x\,\mathrm{d}W$ with $\mathcal{L}v = \frac{1}{2}a^2 v_{xx} + bv_x + \dot{v}$. Supermartingale in general and martingale at the optimum suggest

$$\sup_u\Big\{K(t,x,u) + \tfrac{1}{2}a(t,x,u)^2 V_{xx} + b(t,x,u)V_x + \dot{V}\Big\} = 0, \qquad U^* = \arg\max_u\{K + \mathcal{L}V\}$$

The HJB only produces a *candidate*; conditions (i) to (iv) must still be checked.

Minimisation Version: for a cost to be minimised, the inequalities flip: M^U must be a *submartingale* for every control, a martingale for U^*, with $\limsup_t \mathbb{E}[V(t,X_t^U)] \leq 0$ and $\lim_t \mathbb{E}[V(t,X_t^{U^*})] = 0$; then $\inf_U J = V(0,x)$. The HJB uses inf instead of sup, as in the flypaper problem below.

9.3 Whittle's Flypaper Problem

Problem: $X_t = x + B_t + \int_0^t U_s\,\mathrm{d}s$ on $(0,z)$, exit time τ from $(0,z)$, minimise $\mathbb{E}\big[\int_0^\tau \frac{1}{2}(1+U_s^2)\,\mathrm{d}s + k(X_\tau)\big]$ with given $k(0), k(z) > 0$: steering costs effort, but waiting also costs time.

HJB: look for G with $G(0) = k(0)$, $G(z) = k(z)$ and $\inf_u\{\frac{1}{2}G'' + uG' + \frac{1}{2}(1+u^2)\} = 0$. The minimiser is $u = -G'$, so

$$G'' - (G')^2 + 1 = 0, \qquad U_t^* = -G'(X_t)$$

Linearisation: $g = e^{-G}$ satisfies $g'' = ((G')^2 - G'')g = g$, so $g = A_0\cosh x + A_1 \sinh x$ with

$$A_0 = e^{-k(0)}, \qquad A_1 = e^{-k(z)}\operatorname{cosech} z - e^{-k(0)}\coth z, \qquad G = -\ln g$$

Verification: G' is bounded on $[0,z]$, so under U^*, M^* is a nonnegative local martingale with $\mathbb{E}[\tau^*] < \infty$, hence an L^2 bounded (uniformly integrable) martingale and $G(x) =$ cost of U^*. For any other control with finite expected cost, completing the square gives $M_t^U \geq G(x) + N_t$ with N a martingale, so every control costs at least $G(x)$.

9.4 Perpetual Merton Problem

Candidate: time homogeneity gives $V(t,x) = e^{-\delta t}H(x)$, and scaling ($(\pi, C)$ admissible from x implies $(\pi, \varsigma C)$ admissible from ςx) gives $H(\varsigma x) = \varsigma^{1-R}H(x)$, so $V = Ae^{-\delta t}\frac{x^{1-R}}{1-R}$. The HJB

$$\sup_{\pi,c}\Big\{e^{-\delta t}\frac{c^{1-R}}{1-R} + \tfrac{1}{2}\pi^2\sigma^2 x^2 V_{xx} + [x(r + \varrho\sigma\pi) - c]V_x + \dot{V}\Big\} = 0$$

yields $c = A^{-1/R}x$ and $\pi = \varrho/(\sigma R)$. Substituting back gives $A = \eta^{-R}$ where

$$\eta = \frac{1}{R}\left(\delta - r(1-R) - \frac{\varrho^2(1-R)}{2R}\right)$$

Solution (if $\eta > 0$):

$$\hat{\pi} = \frac{\mu - r}{\sigma^2 R}, \qquad \hat{C}_t = \eta X_t, \qquad \hat{V}(x) = \eta^{-R}\frac{x^{1-R}}{1-R}$$

Invest the **Merton fraction** and consume a constant fraction η of wealth. The condition $\eta > 0$ is the well posedness condition.

Constant Proportional Strategies: for $\pi_t = \pi$ and $C_t = \varsigma X_t$, wealth is a geometric Brownian motion and

$$J = \frac{x^{1-R}}{1-R}\frac{\varsigma^{1-R}}{F(\pi,\varsigma)}, \qquad F(\pi,\varsigma) = \delta - (1-R)\left(r + \varrho\sigma\pi - \frac{\pi^2\sigma^2 R}{2} - \varsigma\right)$$

provided $F > 0$. Since $F(\hat{\pi}, \eta) = \eta$, the candidate attains $\hat{V}(x)$.

Comparative Statics of η: $\partial\eta/\partial\delta = 1/R > 0$, so a more impatient agent consumes more. The effects of r and ϱ^2 carry the sign of $-(1-R)$: for $R < 1$ better investment opportunities lower the consumption rate (substitution dominates), for $R > 1$ they raise it (income effect dominates). In risk aversion,

$$\frac{\partial\eta}{\partial R} = -\frac{\delta - r + \varrho^2/2}{R^2} + \frac{\varrho^2}{R^3}$$

so if $r < \delta + \varrho^2/2$, η first rises and then falls, with a maximum at $R = \varrho^2/(\delta - r + \varrho^2/2)$: the consumption rate is *not* monotonic in R.

Verification under Fiat Conditions: call (π, C) fiat admissible if (P) wealth stays strictly positive, (M) the local martingale $\int_0^\cdot e^{-\delta t}\sigma\pi_t X_t^{1-R}\,\mathrm{d}W_t$ is a supermartingale, and (T) $\liminf_t \mathbb{E}[e^{-\delta t}X_t^{1-R}/(1-R)] \geq 0$. If $\eta > 0$, the candidate is optimal among all fiat admissible strategies: Itô on $M_t = \int_0^t e^{-\delta s}\frac{C_s^{1-R}}{1-R}\,\mathrm{d}s + e^{-\delta t}\hat{V}(X_t)$ gives a drift ≤ 0 (because $\hat{V}$ solves the HJB, with equality at the candidate), so $M_t \leq \hat{V}(x) + N_t$; (M) and (T) then give $J(C) \leq \hat{V}(x)$.

9.5 Log Utility in Continuous Time

Maximise $\mathbb{E}\int_0^\infty e^{-\delta s}\ln C_s\,\mathrm{d}s$ with the guess $V = e^{-\delta t}(A\ln x + B)$. Writing $C_t = \varsigma_t X_t$, the drift of M is $e^{-\delta t}$ times

$$(1-\delta A)\ln X_t + (\ln\varsigma - A\varsigma) - \delta B + A\left(r + \varrho\sigma\pi - \tfrac{1}{2}\sigma^2\pi^2\right)$$

Removing the $\ln X$ term forces $A = 1/\delta$; maximising gives

$$\hat{C}_t = \delta X_t, \qquad \hat{\pi} = \frac{\mu - r}{\sigma^2}, \qquad B = \frac{1}{\delta}\left(\ln\delta - 1 + \frac{r}{\delta} + \frac{\varrho^2}{2\delta}\right)$$

which is the CRRA solution at $R = 1$ (then $\eta = \delta$). The log investor holds the myopic growth optimal portfolio, independent of patience. The difficulty in verifying this candidate is that ln is unbounded in both directions, so neither transversality nor the supermartingale property of M comes for free: one must show $\mathbb{E}[e^{-\delta t}\ln X_t] \to 0$ along the candidate and control $\limsup_t \mathbb{E}[e^{-\delta t}\ln X_t]$ for arbitrary admissible strategies.

9.6 The Dual (Martingale) Approach

State Price Density:

$$\zeta_t = e^{-rt}\mathcal{E}(-\varrho W)_t = \exp\left(-\varrho W_t - \left(r + \tfrac{1}{2}\varrho^2\right)t\right), \qquad \mathrm{d}\zeta_t = -\zeta_t(r\,\mathrm{d}t + \varrho\,\mathrm{d}W_t)$$

Budget Constraint: a nonnegative consumption stream is financeable from x if and only if

$$\mathbb{E}\int_0^\infty \zeta_s C_s\,\mathrm{d}s \leq x$$

Necessity: $Y_t = \zeta_t X_t + \int_0^t \zeta_s C_s\,\mathrm{d}s$ has $\mathrm{d}Y_t = \zeta_t X_t(\pi_t\sigma - \varrho)\,\mathrm{d}W_t$, a nonnegative local martingale and hence a supermartingale, so Fatou gives the bound. Sufficiency uses martingale representation. The dynamic problem becomes a static one with a single constraint.

Recipe:

1. **First order condition:** $e^{-\delta t}U'(\hat{C}_t) = \gamma\zeta_t$ for a multiplier $\gamma > 0$
2. **Invert:** $\hat{C}_t = I(\gamma e^{\delta t}\zeta_t)$ with $I = (U')^{-1}$
3. **Budget:** choose $\gamma = \gamma(x)$ so that $\mathbb{E}\int_0^\infty \zeta_t\hat{C}_t\,\mathrm{d}t = x$
4. **Optimality:** concavity gives $U(b) \leq U(a) + (b-a)U'(a)$, so for any feasible C, $J(C) \leq J(\hat{C}) + \gamma\mathbb{E}\int\zeta_t(C_t - \hat{C}_t)\,\mathrm{d}t \leq J(\hat{C})$
5. **Value:** $V'(x) = \gamma(x)$; integrate to recover V

CRRA Check: $I(y) = y^{-1/R}$, so $\hat{C}_t = \gamma^{-1/R}e^{-\delta t/R}\zeta_t^{-1/R}$ and $\mathbb{E}[\zeta_t\hat{C}_t] = \gamma^{-1/R}e^{-\eta t}$. The budget gives $\gamma^{-1/R} = \eta x$, hence $\hat{C}_0 = \eta x$, $V'(x) = \eta^{-R}x^{-R}$ and $V = \eta^{-R}\frac{x^{1-R}}{1-R}$, as before. The dual route is especially useful for non CRRA utilities, where guessing V is hard.

9.7 Summary

Setting	Value	Consumption	Investment
CRRA, 1 period	$\Xi\frac{x^{1-R}}{1-R}$	$\alpha^* x$	λ^* of $(x-c^*)(1+r)$
CRRA, N periods	$\Xi_k\frac{x^{1-R}}{1-R}$	$\alpha_k^* X_k$	λ^*, constant
CRRA, ∞ horizon	$A(1-\tilde{\theta})^{-R}\frac{x^{1-R}}{1-R}$	$(1-\tilde{\theta})X_k$	λ^*, constant
Log, N periods	$\alpha_k\ln x + \beta_k$	$\frac{A_k}{\alpha_k}X_k$	ϕ^*, constant
Merton, CRRA	$\eta^{-R}\frac{x^{1-R}}{1-R}$	ηX_t	$\frac{\mu-r}{\sigma^2 R}$
Merton, log	$\frac{1}{\delta}\ln x + B$	δX_t	$\frac{\mu-r}{\sigma^2}$
Whittle	$G = -\ln g$		$-G'(X_t)$

10 BSDEs and European Options

Throughout we work in continuous time on $[0,T]$ with a one dimensional Brownian motion W and its natural filtration; (in)equalities between random variables hold almost surely. The key spaces are

$$\mathbb{H}^2 = \Big\{\text{predictable } X : \mathbb{E}\int_0^T X_s^2\,\mathrm{d}s < \infty\Big\}, \quad \mathcal{S}^2 = \Big\{\text{adapted } X : \mathbb{E}\big[\sup_{t\leq T} X_t^2\big] < \infty\Big\}$$

and $L^2(\mathcal{F}_T)$, the square integrable $\mathcal{F}_T$ measurable payoffs.

10.1 Wealth as a Forward SDE

Generalised Black Scholes: $\mathrm{d}B_t = r_t B_t\,\mathrm{d}t$ and $\mathrm{d}S_t = S_t(\mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t)$, with r, μ, σ predictable, bounded and $\sigma > 0$ (constants give the classical model). With $\varphi_t \in \mathbb{H}^2$ the *amount* held in the stock, the self financing condition

$$\mathrm{d}V_t = \frac{\varphi_t}{S_t}\,\mathrm{d}S_t + \frac{V_t - \varphi_t}{B_t}\,\mathrm{d}B_t$$

says that wealth changes only through price moves, with no external cash added or withdrawn. Substituting and setting $Z_t = \varphi_t\sigma_t$, $\theta_t = (\mu_t - r_t)/\sigma_t$ (market price of risk):

$$-\mathrm{d}V_t = f(t,V_t,Z_t)\,\mathrm{d}t - Z_t\,\mathrm{d}W_t, \qquad f^{lin}(t,v,z) = -r_t v - \theta_t z$$

Given (x,φ) this is a forward SDE whose solution lies in $\mathcal{S}^2$ with continuous paths.

10.2 Replication as a BSDE

To replicate a European payoff $\eta \in L^2(\mathcal{F}_T)$ we fix the *terminal* value instead and look for $(V,Z) \in \mathcal{S}^2 \times \mathbb{H}^2$ with

$$V_t = \eta + \int_t^T f(s,V_s,Z_s)\,\mathrm{d}s - \int_t^T Z_s\,\mathrm{d}W_s$$

a **BSDE** with **driver** f and **terminal condition** η. The price is V_t and the hedge is $\varphi_t = Z_t/\sigma_t$. The unknown Z is what keeps V adapted while hitting a random target.

Linear BSDE: for f^{lin} there is a unique solution and

$$V_t = \mathbb{E}\left[\frac{\Gamma_T}{\Gamma_t}\eta\,\middle|\,\mathcal{F}_t\right], \qquad \mathrm{d}\Gamma_t = -\Gamma_t(r_t\,\mathrm{d}t + \theta_t\,\mathrm{d}W_t),\ \Gamma_0 = 1$$

Proof idea: Itô on $\Gamma_t V_t$ shows it is a martingale ($\mathrm{d}(\Gamma V) = \Gamma(Z - \theta V)\,\mathrm{d}W$), and Z comes from the martingale representation of ΓV. Since $\Gamma_t = e^{-\int_0^t r_s\,\mathrm{d}s}\mathcal{E}(-\theta\cdot W)_t$, a Girsanov change of measure gives the familiar risk neutral formula

$$V_t = \mathbb{E}_{\mathbb{Q}}\left[e^{-\int_t^T r_s\,\mathrm{d}s}\eta\,\middle|\,\mathcal{F}_t\right], \qquad \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\Big|_{\mathcal{F}_t} = \mathcal{E}\Big(-\int_0^\cdot \theta_s\,\mathrm{d}W_s\Big)_t$$

Affine Driver: for $f = \alpha_t + \beta_t y + \gamma_t z$ with bounded predictable coefficients,

$$V_t = \mathbb{E}\left[\Gamma_{t,T}\eta + \int_t^T \Gamma_{t,s}\alpha_s\,\mathrm{d}s\,\middle|\,\mathcal{F}_t\right], \qquad \mathrm{d}\Gamma_{t,s} = \Gamma_{t,s}(\beta_s\,\mathrm{d}s + \gamma_s\,\mathrm{d}W_s),\ \Gamma_{t,t} = 1$$

The classical and generalised Black Scholes BSDEs are the special cases $\alpha = 0$, $\beta = -r$, $\gamma = -\theta$. The proof template is always the same: show that $\Gamma_t Y_t + \int_0^t \Gamma_s\alpha_s\,\mathrm{d}s$ is a square integrable martingale, read off Y by taking conditional expectations, obtain Z from martingale representation, and conversely check that this candidate solves the equation.

The Simplest Case: with $f = 0$, taking $\mathbb{E}(\cdot \mid \mathcal{F}_t)$ in $V_t = \eta - \int_t^T Z_s\,\mathrm{d}W_s$ gives $V_t = \mathbb{E}(\eta \mid \mathcal{F}_t)$, and Z is the integrand in the martingale representation of this

martingale. Every linear BSDE is this case in disguise after discounting and a change of measure; in particular $e^{-rt}Z_t$ is the integrand of the $\mathbb{Q}$ martingale $e^{-rt}V_t$.

Markov Case and PDE: if r, μ, σ are deterministic and $\eta = h(S_T)$, then $V_t = u(t, S_t)$. Matching the Itô expansion of $u(t, S_t)$ with the BSDE gives $Z_t = \sigma_t S_t \partial_s u$ (the hedge holds $\partial_s u$ shares, the delta) and the Black Scholes PDE

$$\partial_t u + r_t s\, \partial_s u + \tfrac{1}{2}\sigma_t^2 s^2 \partial_{ss} u - r_t u = 0, \qquad u(T, s) = h(s)$$

10.3 An Imperfect Market: Different Rates

Suppose cash can be lent at r_t but borrowed at $R_t \geq r_t$. Positive cash $(V - \varphi)^+$ earns r and a shortfall $(V - \varphi)^-$ costs R, so

$$\mathrm{d}V_t = \Big[r_t V_t + \theta_t Z_t - (R_t - r_t)\Big(V_t - \frac{Z_t}{\sigma_t}\Big)^- \Big]\,\mathrm{d}t + Z_t\,\mathrm{d}W_t$$

and the pricing BSDE has the **non linear** driver

$$f(t, y, z) = -r_t y - \theta_t z + (R_t - r_t)\Big(y - \frac{z}{\sigma_t}\Big)^-$$

It is Lipschitz and **convex** in (y, z), since $x \mapsto x^-$ is convex. Prices are no longer linear in the payoff: the seller's price of a portfolio of claims can be below the sum of separate prices, because borrowing needs may net out.

10.4 General Theory

Lipschitz Driver: $f(\omega, t, y, z)$ is predictable in (ω, t) and Borel in (y, z), satisfies $\mathbb{E}\int_0^T |f(t, 0, 0)|^2\,\mathrm{d}t < \infty$, and there is $K \geq 0$ with

$$|f(t, y_1, z_1) - f(t, y_2, z_2)| \leq K\big(|y_1 - y_2| + |z_1 - z_2|\big)$$

A pair (f, η) with f Lipschitz and $\eta \in L^2(\mathcal{F}_T)$ is called *standard*.

Existence and Uniqueness (Pardoux and Peng, 1990): for standard parameters the BSDE has a unique solution $(V, Z) \in \mathcal{S}^2 \times \mathbb{H}^2$. The proof uses a contraction (Banach fixed point) in weighted norms $\mathbb{E}\int_0^T e^{\beta s}|\cdot|^2\,\mathrm{d}s$ with β large.

Comparison Theorem: if $\eta^1 \geq \eta^2$ and $f^1(t, V_t^2, Z_t^2) \geq f^2(t, V_t^2, Z_t^2)$ $\mathrm{d}t \otimes \mathrm{d}\mathbb{P}$ a.e., then $V_t^1 \geq V_t^2$ for all t. A larger payoff or a more costly driver can never have a lower price.

Random Horizon: for a stopping time $\tau \leq T$ and $\eta \in L^2(\mathcal{F}_\tau)$, solve with driver $f(t, y, z)\mathbb{1}_{\{t \leq \tau\}}$ on $[0, T]$; then $V_t = \eta$ on $\{t \geq \tau\}$.

10.5 Non Linear Pricing Operators

The f **evaluation** (conditional f expectation) maps a payoff to the first component of the BSDE solution:

$$\mathcal{E}^f_{t,T} : L^2(\mathcal{F}_T) \to L^2(\mathcal{F}_t), \qquad \eta \mapsto V_t$$

For $f = 0$ it is $\mathbb{E}(\cdot \mid \mathcal{F}_t)$; for $f = f^{lin}$ it is the risk neutral price. For stopping times $S \leq \nu \leq \tau$ it satisfies:

- $\mathcal{E}^f_{S,S}(\eta) = \eta$ for $\eta \in L^2(\mathcal{F}_S)$
- **Monotonicity:** $\eta^1 \leq \eta^2 \Rightarrow \mathcal{E}^f_{S,\tau}(\eta^1) \leq \mathcal{E}^f_{S,\tau}(\eta^2)$ (from comparison)
- **Time consistency:** $\mathcal{E}^f_{S,\nu}\big(\mathcal{E}^f_{\nu,\tau}(\eta)\big) = \mathcal{E}^f_{S,\tau}(\eta)$
- **Zero one law:** if $f(t, 0, 0) = 0$, then $\mathbb{1}_A \mathcal{E}^f_{S,T}(\eta) = \mathcal{E}^f_{S,T}(\mathbb{1}_A \eta)$ for $A \in \mathcal{F}_S$
- **Continuity** in the terminal time and terminal condition
- **Convexity:** a driver convex in (y, z) gives an operator convex in η

Proving Such Properties: most follow from uniqueness. For instance, linearity of the Black Scholes pricing operator: if (V, Z) and (V', Z') solve the linear BSDE for η and η', then by linearity of f^{lin} the pair $(V + V', Z + Z')$ solves it for $\eta + \eta'$, and (cV, cZ) solves it for $c\eta$; uniqueness then gives $\mathcal{E}^{f^{lin}}_{t,T}(c\eta + \eta') = c\,\mathcal{E}^{f^{lin}}_{t,T}(\eta) + \mathcal{E}^{f^{lin}}_{t,T}(\eta')$. The same argument fails for a non linear driver, which is exactly why prices become non linear. Such operators also describe recursive utilities and dynamic risk measures. Beyond the Brownian setting, martingale representation is replaced by a predictable representation property, and comparison may need extra conditions on the driver when there are jumps.

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11 American Options and Reflected BSDEs

11.1 Superhedging

An American option pays ξ_τ if exercised at the stopping time $\tau \in \mathcal{T}_0$ (stopping times valued in $[0, T]$), where ξ is adapted, RCLL and in $\mathcal{S}^2$; for simplicity we take ξ continuous. A **superhedge** from capital x is a self financing $\varphi \in \mathbb{H}^2$ with

$$V_\tau^{x,\varphi} \geq \xi_\tau \ \ \forall \tau \in \mathcal{T}_0 \quad \Longleftrightarrow \quad V_t^{x,\varphi} \geq \xi_t \ \ \forall t \in [0, T] \ \text{ a.s.}$$

("for all t" sits *inside* the almost sure statement). The **seller's superhedging price** is $u_0 = \inf\{x : \text{a superhedge from } x \text{ exists}\}$.

Linear Case: taking $r = 0$ (or discounting) and $\mu = r$ (or changing measure) so that $\mathbb{P} = \mathbb{Q}$,

$$u_0 = \sup_{\tau \in \mathcal{T}_0} \mathbb{E}(\xi_\tau) = Y_0$$

where Y solves the reflected BSDE below with driver 0. The first equality is a **pricing hedging duality**: every wealth process is then a martingale, so $x = \mathbb{E}V_\tau \geq \mathbb{E}\xi_\tau$ gives $u_0 \geq \sup_\tau \mathbb{E}\xi_\tau$; conversely $\varphi_t^* = Z_t/\sigma_t$ from the reflected BSDE superhedges from Y_0.

11.2 Tools

Essential Supremum: for an uncountable family $(X_i)_{i \in I}$ the pointwise supremum may not be measurable. The essential supremum is the a.s. smallest random variable dominating every X_i, and it equals the supremum over a countable subfamily. If the family is **directed upwards** (for any i, j some k has $X_k \geq X_i \vee X_j$), it is the a.s. limit of a nondecreasing sequence X_{i_n}.

Doob Meyer Decomposition: a right continuous supermartingale $Y \in \mathcal{S}^2$ decomposes uniquely as $Y_t = M_t - A_t$ with M a square integrable martingale and A predictable, RCLL, nondecreasing, $A_0 = 0$. In a Brownian filtration $M_t = M_0 + \int_0^t Z_u\,\mathrm{d}W_u$ with $Z \in \mathbb{H}^2$; if Y is continuous, so is A.

Optional Sampling: for a right continuous supermartingale and bounded stopping times $S \leq \tau$, $\mathbb{E}(Y_\tau \mid \mathcal{F}_S) \leq Y_S$.

11.3 Optimal Stopping and the Snell Envelope

The value family is $v(S) = \operatorname{ess\,sup}_{\tau \in \mathcal{T}_S} \mathbb{E}(\xi_\tau \mid \mathcal{F}_S)$ for $S \in \mathcal{T}_0$, with $\mathcal{T}_S$ the stopping times between S and T.

Construction in Eight Steps:

1. The family $v(S)$ is admissible and square integrable
2. It is **aggregated** by a unique RCLL process (v_t) with $v_S = v(S)$, continuous when ξ is
3. (v_t) is a supermartingale dominating ξ (the family $\mathbb{E}(\xi_\tau \mid \mathcal{F}_S)$ is directed upwards)
4. **Snell envelope:** (v_t) is the *smallest* supermartingale dominating ξ
5. Doob Meyer: $v_t = M_t - A_t$
6. Martingale representation: $v_t = v_0 + \int_0^t Z_u\,\mathrm{d}W_u - A_t$ with A continuous
7. With $\tau_S^* = \inf\{t \geq S : v_t = \xi_t\}$, v is a martingale on $[S, \tau_S^*]$ (proved through $\tau_S^\lambda = \inf\{t \geq S : \lambda v_t \leq \xi_t\}$, $\lambda \uparrow 1$, Maingueneau's argument)
8. A only increases when v touches the obstacle: $\int_0^T \mathbb{1}_{\{v_t > \xi_t\}}\,\mathrm{d}A_t = 0$

Hence τ_0^*, the first time the value meets the payoff, is an optimal exercise time.

11.4 Reflected BSDE

Definition: $(Y, Z, A) \in \mathcal{S}^2 \times \mathbb{H}^2 \times \mathcal{S}^2$ solves the RBSDE with driver f and lower obstacle ξ if

$$Y_t = \xi_T + \int_t^T f(s, Y_s, Z_s)\,\mathrm{d}s - \int_t^T Z_s\,\mathrm{d}W_s + A_T - A_t, \qquad Y_t \geq \xi_t$$

with A continuous, predictable, nondecreasing, $A_0 = 0$, and the **Skorokhod (minimality) condition** $\int_0^T \mathbb{1}_{\{Y_t > \xi_t\}}\,\mathrm{d}A_t = 0$. The process A is the minimal push needed to keep Y above the obstacle.

Running Reward: for a driver process $f \in \mathbb{H}^2$ (not depending on Y, Z), $v_t + \int_0^t f\,\mathrm{d}u$ is the Snell envelope of $\xi_t + \int_0^t f\,\mathrm{d}u$, and the eight steps give existence. For uniqueness, take any solution: A is constant on $[S, \tau_S^*]$, so $Y + \int_0^\cdot f$ is a martingale there and

$$Y_S = \mathbb{E}\Big[\xi_{\tau_S^*} + \int_S^{\tau_S^*} f(u)\,\mathrm{d}u \,\Big|\, \mathcal{F}_S\Big] \leq v(S)$$

while $Y + \int_0^\cdot f$ is a supermartingale above the obstacle, hence above the Snell envelope; so $Y_S = v(S)$. Uniqueness also follows from a priori estimates in weighted norms.

11.5 The Non Linear Case

With a general Lipschitz driver (e.g. different borrowing and lending rates), define the **non linear optimal stopping** value

$$v(S) = \operatorname*{ess\,sup}_{\tau \in \mathcal{T}_S} \mathcal{E}^f_{S,\tau}(\xi_\tau)$$

aggregated into a continuous process (v_t).

Existence and Uniqueness: the RBSDE with Lipschitz f and continuous obstacle $\xi \in \mathcal{S}^2$ has a unique solution. Freeze (y, z) in the driver, solve the running reward RBSDE, and show the resulting map $\Phi(y, z) = (Y, Z)$ is a strict contraction in β weighted norms for β large.

Main Theorem:

$$u_0 = \sup_{\tau \in \mathcal{T}_0} \mathcal{E}^f_{0,\tau}(\xi_\tau) = Y_0$$

The superhedging price in the imperfect market, the value of the non linear optimal stopping problem and the RBSDE solution coincide, and $\tau^* = \inf\{t : Y_t = \xi_t\}$ is optimal. For an RCLL (not continuous) payoff the price characterisation survives but an optimal stopping time may not exist.

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12 Exponential Utility via BSDEs

12.1 Problem

Take $r = 0$ (so $B \equiv 1$), $\mathrm{d}S_t = S_t(\mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t)$ with bounded predictable coefficients, $\sigma > 0$, and a bounded liability η. With φ the amount in the stock, $V_t^{x,\varphi} = x + \int_0^t \varphi_s\,\mathrm{d}S_s/S_s$, and we seek

$$v(x) = \sup_{\varphi \in \mathcal{A}(x)} \mathbb{E}\big[U(V_T^{x,\varphi} - \eta)\big], \qquad U(x) = -e^{-\gamma x},\ \gamma > 0$$

This is utility maximisation with a claim to deliver, not replication.

12.2 Martingale Optimality Principle

Build processes J^φ with (1) $J_T^\varphi = U(V_T^{x,\varphi} - \eta)$, (2) $J_0^\varphi = C$ the same constant for all φ, (3a) J^φ a supermartingale for every φ, and (3b) $J^{\hat\varphi}$ a martingale for some $\hat\varphi$. Then $\mathbb{E}U(V_T^\varphi - \eta) \leq C = \mathbb{E}U(V_T^{\hat\varphi} - \eta)$, so $\hat\varphi$ is optimal and $v(x) = C$.

Ansatz: $J_t = U(V_t^{x,\varphi} - Y_t)$ where Y solves an auxiliary BSDE $Y_t = \eta + \int_t^T f(s, Z_s)\,\mathrm{d}s - \int_t^T Z_s\,\mathrm{d}W_s$ whose driver f must not depend on φ. Conditions (1) and (2) hold automatically, with $C = U(x - Y_0)$.

12.3 Finding the Driver

Substituting the dynamics,

$$J_t = -e^{-\gamma(x - Y_0)} \exp\Big(-\gamma \int_0^t (\varphi_s \mu_s + f(s, Z_s))\,\mathrm{d}s - \gamma \int_0^t (\varphi_s \sigma_s - Z_s)\,\mathrm{d}W_s\Big)$$

Completing the exponential martingale, $J_t = e^{-\gamma(x - Y_0)} M_t C_t$ with

$$M_t = \mathcal{E}\Big(-\gamma \int_0^\cdot (\varphi_s \sigma_s - Z_s)\,\mathrm{d}W_s\Big)_t, \qquad C_t = U\Big(\int_0^t \psi_s\,\mathrm{d}s\Big)$$

$$\psi_s = \varphi_s \mu_s + f(s, Z_s) - \tfrac{\gamma}{2}(\varphi_s \sigma_s - Z_s)^2$$

M is a positive (local) martingale. A nonnegative martingale times a nonincreasing nonpositive process is a supermartingale, so we need $\psi \geq 0$ for every φ and $\psi = 0$ for the optimiser.

Pointwise Maximisation: with $\theta_s = \mu_s/\sigma_s$ and $p = \varphi_s \sigma_s$, ψ is a concave quadratic in p, maximised at $p = Z_s + \theta_s/\gamma$ with maximum $f(s, Z_s) + \theta_s Z_s + \theta_s^2/(2\gamma)$. Setting this maximum to zero:

$$f(s, z) = -\theta_s z - \frac{\theta_s^2}{2\gamma}, \qquad \hat\varphi_s = \frac{\mu_s}{\gamma \sigma_s^2} + \frac{Z_s}{\sigma_s}$$

Interpretation: the optimal position is the Merton type term $\mu/(\gamma\sigma^2)$ (here in money, reflecting constant *absolute* risk aversion) plus a hedge Z/σ of the liability. The value is

$$v(x) = -e^{-\gamma(x - Y_0)}$$

so Y_0 acts as the certainty equivalent cost of the liability. The driver is affine, hence Lipschitz, and the affine formula gives explicitly

$$Y_0 = \mathbb{E}_{\mathbb{Q}}\Big[\eta - \int_0^T \frac{\theta_s^2}{2\gamma}\,\mathrm{d}s\Big]$$

with $\mathbb{Q}$ the martingale measure (density $\mathcal{E}(-\theta \cdot W)$). In incomplete markets or for other utilities the driver is typically **quadratic** in z and needs quadratic BSDE theory. A remaining technical step is to check that $M_t C_t$ is a true supermartingale, not only a local one.